

Project 2: Extending the Hello World Example to an n-Qubit GHZ State

Project Goal:

In this project, we are trying to prepare and analyze the n-qubit GHZ (Greenberger-Horne-Zeilinger) state using Qiskit. The GHZ states help us to understand the limits of Quantum phenomena and to understand better about quantum entanglement. The GHZ state also demonstrates how quantum systems can exhibit correlations that cannot be explained classically. In this project, we ran the GHZ circuit on real IBM Quantum Backends, and compared the results to the output.

Approach:

Firstly, we built the GHZ circuit by applying a Hadamard gate on the first qubit, creating a superposition of 0 and 1. Then, we added a series of CNOT gates entangle the remaining qubits with the first one.

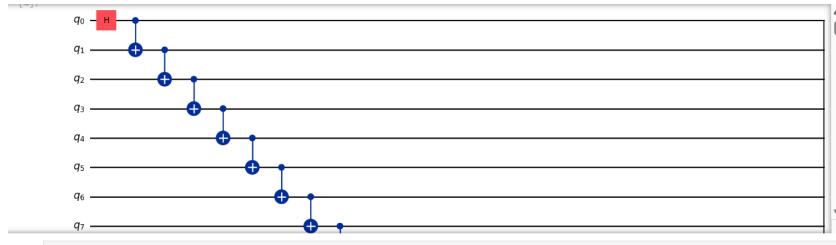


Figure 1: Part of the 100 qubit GHZ circuit

After constructing the circuit and drawing it out for n-qubits, which is in our case was $n = 100$, we then did the operators. We then look at the ZZ operators to map out how our expectation values are more affected by the noise.

Then, we transpiled the circuit into two IBM hardware devices - one of them is a smaller backend of 133 qubits (ibm_torino), the other one is a bit larger of 156 qubits (ibm_marrakesh).

Transpilation helps in the circuit to adapt and run on the hardware's available qubit connection and setup. This may lead to reduction in the noise and errors that may be caused to our expected value.

Results & observations:

To proceed in using the hardware, we first checked the hardware that were connected to our system. It was found to be 'ibm_fez,' 'ibm_torino,' and 'ibm_marrakesh.'

Firstly, we connected the "ibm_torino" hardware and it gave us the following result on the graph for 100 qubits:

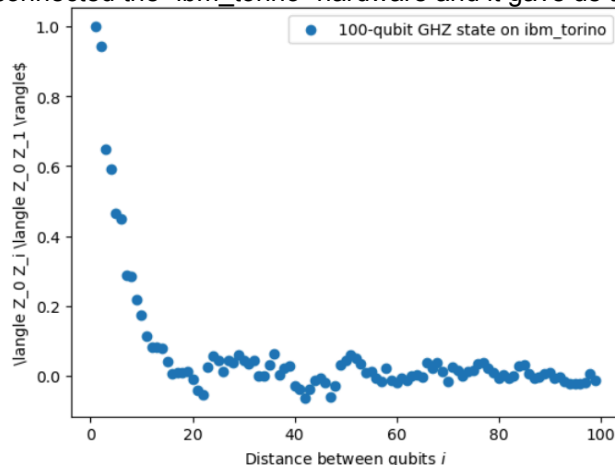


Figure 2: Result on ibm_torino

Next, we attempted to connect the larger backend such as the "ibm_marrakesh," it provided us with the result as the following:

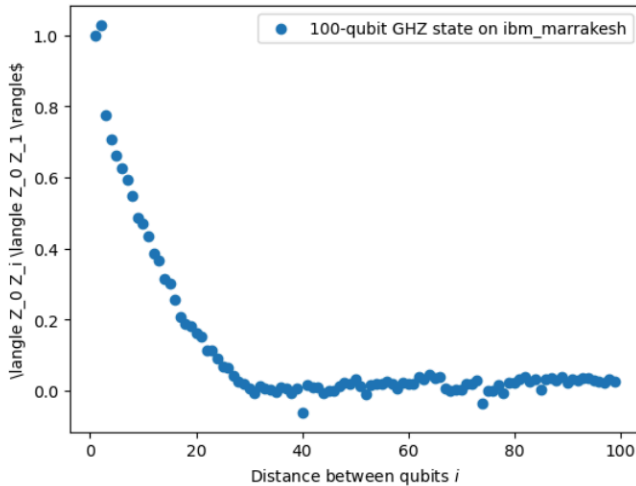


Figure 3: Result on ibm_marrakesh

Transpilation optimized the circuit layout to fit each device's qubit connectivity, helping reduce gate errors and improve performance.

Expected VS. Observed Result:

Expectation: In an ideal GHZ circuit, all the qubits should be correlated. Hence, the expectation value of the two-qubit correlation operator ZZ should remain close to 1 for all qubits i , regardless of their distance.

Observed result:

The GHZ circuit was initially transpiled and executed on ibm_torino.

The resulting correlation plot (Figure 2) shows strong short-range correlations near the reference qubit (close to 1.0) that quickly decay as qubit distance increases.

The same 100-qubit GHZ circuit was then executed on ibm_marrakesh.

The resulting plot (Figure 3) shows a smoother and slightly higher correlation range compared to ibm_torino, indicating that ibm_marrakesh provided better qubit connectivity and lower overall noise.

Conclusion:

To conclude, we can say that both ibm_torino and ibm_marrakesh began with $\langle Z_0 Z_i \rangle$ values close to 1 near the reference qubit, indicating strong short-range correlations at the start of the circuit. As the distance between qubits increased, these correlations gradually decayed, approaching near-zero values beyond approximately 20 qubits. This decline is primarily caused by hardware noise and decoherence effects inherent in large-scale quantum systems. However, the ibm_marrakesh backend demonstrated slightly better coherence over longer distances compared to ibm_torino, suggesting that it has more stable qubit coupling, improved calibration, and lower overall error rates.

Transpilation:

Transpilation is the process of converting a quantum circuit into a form that is scaled to be used for a specific quantum hardware. It adapts to the gate sets. It also reduces circuit depth and noise by improving accuracy of results on actual quantum computers. In this project transpilation was used for the GHZ circuit to match with the specific hardwares used such as "ibm_torino" and "ibm_fez."

Dynamic Decoupling:

It is a method that is used to reduce decoherence between qubits in the system. It can also help in reducing phase errors. It is a way of inserting sequences of carefully timed pulses during idle periods to protect qubit decoherence. This method is used in hardware of quantum computers to reduce decoherence.